

PL — Polymers and Life

OCR B (Salters) H433 storyline **PL**. Polymers, proteins and DNA: condensation polymers, amines and amides, amino acid and protein chemistry, optical isomerism, enzyme catalysis and structural analysis.

1. Amines and Amino Acids

Amines

Amine: An organic compound derived from ammonia (NH_3) by replacing one or more hydrogen atoms with alkyl or aryl groups. Primary amines have the formula RNH_2 , secondary amines R_2NH , and tertiary amines R_3N .

Preparation of amines:

- Reaction of a halogenoalkane with excess ammonia in a sealed tube at elevated temperature: $\text{RBr} + 2\text{NH}_3 \rightarrow \text{RNH}_2 + \text{NH}_4\text{Br}$. Excess ammonia is used to minimise further substitution.
- Reduction of a nitrile with LiAlH_4 in dry ether or by catalytic hydrogenation: $\text{RCN} + 4[\text{H}] \rightarrow \text{RCH}_2\text{NH}_2$.
- Reduction of a nitro compound (important for aromatic amines): $\text{C}_6\text{H}_5\text{NO}_2 + 6[\text{H}] \rightarrow (\text{Sn/conc. HCl, reflux}) \text{C}_6\text{H}_5\text{NH}_2 + 2\text{H}_2\text{O}$.

Base properties of amines: Amines act as Brønsted-Lowry bases because the lone pair on nitrogen can accept a proton: $\text{RNH}_2 + \text{H}^+ \rightarrow \text{RNH}_3^+$. Aliphatic amines are stronger bases than ammonia because the electron-donating alkyl groups increase electron density on the nitrogen, making the lone pair more available for protonation. Aromatic amines (e.g. phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$) are weaker bases than ammonia because the nitrogen lone pair is delocalised into the benzene ring, reducing its availability.

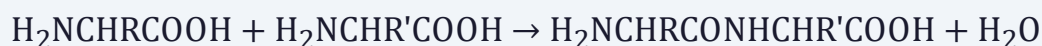
Amino Acids

Amino acid: An organic molecule containing both an amine group ($-\text{NH}_2$) and a carboxylic acid group ($-\text{COOH}$). The α -amino acids found in proteins have both groups attached to the same carbon atom (the α -carbon), along with a hydrogen atom and a variable side chain (R group).

Zwitterions: In solution at the isoelectric point, amino acids exist predominantly as zwitterions — the amine group is protonated ($-\text{NH}_3^+$) and the carboxylic acid group is deprotonated ($-\text{COO}^-$). This gives the molecule both a positive and a negative charge but an overall neutral charge. Zwitterions have high melting points and are soluble in water due to their ionic character.

In acidic solution, the COO^- group is protonated, giving a positive ion. In basic solution, the NH_3^+ group is deprotonated, giving a negative ion.

Peptide Bonds and Proteins



A peptide bond ($-\text{CONH}-$) forms by a condensation reaction between the $-\text{COOH}$ of one amino acid and the $-\text{NH}_2$ of another, releasing water. Proteins are polypeptides formed from many amino acids linked by peptide bonds. Peptide bonds can be broken by hydrolysis (acid or base conditions, or enzymatically).

2. Polymers

Addition Polymers

Formed from alkene monomers by breaking $\text{C}=\text{C}$ double bonds. No by-product is formed. Examples: poly(ethene), poly(propene), PVC, PTFE. They are non-biodegradable and chemically inert (strong $\text{C}-\text{C}$ backbone).

Condensation Polymers

Condensation polymerisation: Monomers join together with the loss of a small molecule (usually water) at each linkage. The monomers must have two functional groups each (they are bifunctional).

Polyesters: Formed from a diol ($\text{HO}-\text{R}-\text{OH}$) and a dicarboxylic acid ($\text{HOOC}-\text{R}'-\text{COOH}$), or from a hydroxycarboxylic acid ($\text{HO}-\text{R}-\text{COOH}$). The linkage is an ester bond ($-\text{COO}-$). Example: Terylene (PET) from ethane-1,2-diol and benzene-1,4-dicarboxylic acid.

Polyamides: Formed from a diamine ($\text{H}_2\text{N}-\text{R}-\text{NH}_2$) and a dicarboxylic acid ($\text{HOOC}-\text{R}'-\text{COOH}$), or from an amino acid. The linkage is a peptide/amide bond ($-\text{CONH}-$). Examples: nylon-6,6 from 1,6-diaminohexane and hexanedioic acid; Kevlar from 1,4-diaminobenzene and benzene-1,4-dicarboxylic acid.

Biodegradability: Condensation polymers are generally biodegradable because the ester or amide bonds can be hydrolysed by water, acids, bases, or enzymes. Addition polymers are generally non-biodegradable. Biodegradable polymers reduce landfill waste but may produce greenhouse gases (CO_2 , CH_4) during decomposition.

Exam Tip: When drawing condensation polymer repeat units, remember to show the linkage (ester or amide bond) and indicate where water was lost. The repeat unit should be enclosed in brackets with continuation bonds on each side. Be able to identify the monomers from a given polymer structure by breaking the ester or amide bonds and adding back H and OH.